LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Metals and Non-metals

Physical properties

Metals	Non-metals
Lustrous, hard, malleable, ductile, sonorous	Dull, brittle (solids), not sonorous
Good conductors of heat and electricity	Poor conductors
Mostly solids with high melting points	Solids, liquid (bromine) or gases

Exceptions: mercury is a liquid metal. Sodium and potassium are soft and can be cut with a knife. Gallium and caesium melt on the palm. Iodine is lustrous. Diamond (carbon) is the hardest natural substance. Graphite conducts electricity.

Chemical properties

- Metal + oxygen → **basic** oxide: $2Cu + O_2 \rightarrow 2CuO$. Non-metal oxides are acidic or neutral.
- **Amphoteric** oxides (Al_2O_3, ZnO) react with both acids and bases: $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$
- With water: Na and K react violently with cold water: $2Na + 2H_2O \rightarrow 2NaOH + H_2$ + heat. Ca reacts less violently. Mg reacts with hot water. Al, Fe and Zn react only with steam: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. Pb, Cu, Ag and Au do not react.
- With dilute acids: metals above hydrogen give H_2. Nitric acid is an oxidising agent, so usually no H_2 (except Mg and Mn with very dilute HNO_3).
- Displacement: a more reactive metal displaces a less reactive one: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

Reactivity series: K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au

Ionic compounds (e.g. NaCl, $MgCl_2$)

- Formed by transfer of electrons. $Na \rightarrow Na^+ + e^-$ and $Cl + e^- \rightarrow Cl^-$
- Hard solids with high melting and boiling points (strong forces between ions).
- Soluble in water, insoluble in kerosene and petrol.
- Conduct electricity when molten or in solution (ions can move), not when solid.

Extraction of metals

Reactivity	Method	Example
Low (Hg, Cu)	Heating alone	$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$; $2HgO \rightarrow 2Hg + O_2$
Middle (Zn, Fe, Pb)	Roasting (sulphide) or calcination (carbonate), then reduction with carbon	$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; $ZnCO_3 \rightarrow ZnO + CO_2$; $ZnO + C \rightarrow Zn + CO$
High (Na, Mg, Ca, Al)	Electrolysis of molten chloride/oxide	Na at cathode, Cl_2 at anode

Thermite reaction: $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + heat (joins railway tracks). **Electrolytic refining:** impure metal is the anode, a thin strip of pure metal is the cathode, and impurities settle as anode mud.

Corrosion & alloys

- Silver turns black (silver sulphide). Copper gets a green coat (basic copper carbonate). Iron rusts (needs both air and water).
- Prevention: painting, oiling, greasing, galvanising (zinc coat), chrome plating, anodising, alloying.

- Alloys: brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn, low melting point), stainless steel (Fe + Ni + Cr). 22 carat gold = 22 parts gold + 2 parts Cu or Ag.

Quick check

1. Why is sodium kept under kerosene?

2. Name a metal that is extracted by electrolysis and one that is obtained by heating alone.

3. Give two conditions needed for iron to rust.